



TECHNICAL MAINTENANCE MANUAL

Fire Alarm Maintenance Manual - UH-FIRE-502

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Client	UrbanHive Facility Technologies Pvt. Ltd.
Site	GreenPark Meridian Residency, Whitefield, Bengaluru
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Review cycle	Annual or after any major incident

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1. Equipment Overview and Technical Specifications

Addressable fire alarm control panel (FACP) serving GreenPark Meridian towers with smoke/heat detection, manual call points, hooter-strobe notification, and integration to lift recall and fire pump.

Model code: UH-FIRE-502

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 230V AC mains with 24V 7Ah battery backup 24h; loop wiring FP200 gold

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-FIRE-502, commissioning checklist CC-033, and spare parts catalogue SP-UH-FIRE-502.

Technicians must verify nameplate data matches AMIS asset record UH-FIRE-502 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

2. Installation Guidelines

Loop wiring in fire-rated conduit. End-of-line resistor 4.7k Ω verified per device. Cause-and-effect matrix uploaded to UrbanHive compliance portal before occupancy certificate.

Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance ≤ 1 ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: panel trouble log review; printer paper check.
- Weekly: random MCP test one per tower; verify hooter audibility at farthest apartment.
- Monthly: smoke detector sensitivity test per 10% sample using test smoke.
- Quarterly: battery discharge test 30 minutes under load.
- Annual: full cause-and-effect test with fire department witness where required.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

Fault 01 - Code F101

Symptom: Loop short circuit trouble

Remediation: Isolate loop section; measure resistance; locate crushed cable in shaft.

Fault 02 - Code F208

Symptom: Detector head contamination alarm

Remediation: Clean chamber with approved aerosol; replace if sensitivity drift > 15%.

Fault 03 - Code F315

Symptom: MCP glass broken nuisance

Remediation: Verify tamper spring; replace element; check for vandalism pattern on floor.

Fault 04 - Code F401

Symptom: Notification appliance circuit open

Remediation: Trace strobe circuit; replace failed 24V sounder on level 12 corridor.

Fault 05 - Code F509

Symptom: Battery backup fail under load

Remediation: Replace battery bank; verify charger float 27.4V; test after 24h recharge.

Fault 06 - Code F602

Symptom: Ground fault on loop 2

Remediation: Megger loop to earth; dry junction boxes in parking level.

Fault 07 - Code F701

Symptom: Printer fault / log not printing

Remediation: Replace thermal paper; clean print head; verify ribbon sensor.

Fault 08 - Code F803

Symptom: BMS fire signal not received

Remediation: Test dry contact output; verify MODBUS mapping point 40021.

Fault 09 - Code F902

Symptom: Lift recall relay not dropping

Remediation: Test relay coil voltage; inspect fire interface wiring diagram UH-FIRE-LIFT-IF.

Fault 10 - Code F1001

Symptom: Repeated isolate events on loop 3

Remediation: Identify device address 247; replace base if intermittent short.

Fault 11 - Code F1104

Symptom: Heat detector rate-of-rise fault

Remediation: Replace detector in kitchenette refuse area; verify spacing compliance.

Fault 12 - Code F1202

Symptom: Panel firmware checksum error

Remediation: Reload approved firmware UH-FACP-2.8.3 from UrbanHive secure portal.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

Part No.	Description	Min stock
SP-F01	Smoke detector head	24
SP-F02	MCP break-glass element	30
SP-F03	24V 7Ah battery	8
SP-F04	Strobe sounder combo	12
SP-F05	End-of-line resistor kit	10
SP-F06	FACP printer paper roll	20

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Life safety system - no bypass without UrbanHive Fire Safety Officer written approval. Maintain detector coverage during any isolate. Report panel trouble to security within 5 minutes.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0033 (08:00-20:00 IST)

Email: support-uh-fire-502@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Fire Alarm` with asset_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-033